CORY L. HOFMANN, MS[1] • DEREK T. HOLYOAK, MS[2] • PAUL M. JURIS, EdD[1,3]

Trunk and Shank Position Influences Patellofemoral Joint Stress in the Lead and Trail Limbs During the Forward Lunge Exercise

STUDY DESIGN: Controlled laboratory study, repeated-measures design.

BACKGROUND: The effects of trunk and shank position on patellofemoral joint stress of the lead limb have been well studied; however, the effects on the trail limb are not well understood.

OBJECTIVES: To test the hypothesis that trunk and shank position may influence patellofemoral joint stress in both limbs during the forward lunge exercise.

METHODS: Patellofemoral kinetics were quantified from 18 healthy participants performing the lunge exercise with different combinations of trunk and shank positions (vertical or forward). A 2-by-3 (limb-by-lunge variation) repeated-measures analysis of variance was performed, using paired *t* tests for post hoc comparisons.

RESULTS: The trail limb experienced greater total patellofemoral joint stress relative to the lead limb, regardless of trunk and shank position ($P<.0001$). The lunge variation with a vertical shank position resulted in significantly greater peak patellofemoral joint stress in the trail limb relative to the lead limb ($P<.0001$). A forward trunk and shank position resulted in the highest patellofemoral stress in the lead limb ($P<.0001$).

CONCLUSION: Trunk and shank positions have a significant influence on patellofemoral joint loading of both limbs during the forward lunge, with the trail limb generally experiencing greater total joint stress. Restricting forward translation of the lead-limb shank may reduce patellofemoral joint stress at the expense of increased stress in the trail limb. Technique recommendations should consider the demands imposed on both knees during this exercise. *J Orthop Sports Phys Ther 2017;47(1):31-40. Epub 4 Nov 2016. doi:10.2519/jospt.2017.6336*

KEY WORDS: *joint moments, knee mechanics, lower extremity, patellofemoral compression, strength training*

The forward lunge exercise is commonly performed in fitness and rehabilitative settings as a resistance exercise for the lower extremity. This exercise can be performed using body weight as resistance or with a variety of exercise implements to add external load and increase demand on the muscles. The forward lunge begins from an anatomically neutral standing position, followed by a unilateral step forward onto the lead limb, a drop in the body's center of mass (as a result of flexion of the hip and knee), and a subsequent return to the starting position. The nonstepping limb, referred to herein as the trail limb, supports body weight during the flight phases of the lead limb and partially supports body weight as weight is transferred to the lead limb. Biomechanical[13,14,16,17] and electromyographical[4,11,12,17] analyses of the lead limb during this exercise and its variants have been performed previously; however, the effects of posture and technique on the mechanics of the trail limb during this exercise have not been reported.

Given the unconstrained nature of the lunge exercise, many variations exist, most of which involve changes in posture. For example, a technique commonly employed during the forward lunge is to incorporate forward lean of the trunk. Trunk position has been shown to have a significant influence on lower extremity joint moments of the lead limb during the lunge exercise,[17] likely due to the fact that the head, arms, and trunk consist of about 55% of body mass.[30] More specifically, a forward lean at the trunk has been reported to significantly increase the ankle plantar flexor moment, hip extensor moment, and activation of gluteus maximus and biceps femoris muscles relative to a vertical trunk posture.[17]

In addition to trunk position, the sagittal plane location of the knee joint center relative to the ankle joint center (shank angle) is another variable that can be modified during the forward lunge exercise. Shank angle has been shown to alter the load distribution between the hip and knee extensors during the bilateral barbell squat.[19] During the lunge, the shank angle can be influenced by

[1]Cybex Research Institute, Medway, MA. [2]Meinig School of Biomedical Engineering, Cornell University, Ithaca, NY. [3]Department of Kinesiology, University of Massachusetts-Amherst, Amherst, MA. No funding was received for this study. Mr Hofmann and Dr Juris are salaried employees of Cybex International, Inc, an exercise equipment manufacturer. This study was approved by the University of Massachusetts-Amherst Institutional Review Board. The authors certify that they have no affiliations with or financial involvement in any organization or entity with a direct financial interest in the subject matter or materials discussed in the article. Address correspondence to Mr Cory Hofmann, Cybex Research Institute, 10 Trotter Drive, Medway, MA 02053. E-mail: chofmann@cybexintl.com Copyright ©2017 *Journal of Orthopaedic & Sports Physical Therapy®*

imposing different step lengths.[14] An investigation of short and long step lengths during the forward lunge, for example, illustrates that stride length has a significant influence on the loading of the cruciate ligaments, the hamstrings, and the quadriceps of the lead limb.[16] It stands to reason that between-limb moments also would be altered as a result of technique variations during exercises such as the forward lunge. However, past biomechanical analyses of this exercise have focused almost exclusively on the lead limb.

One exercise that has been analyzed for differences in lead-limb and trail-limb mechanics is the split squat.[26] The split squat is similar to the forward lunge, but it does not involve stepping forward onto the lead limb or pushing back to return to the starting position. Previous research on the split squat has demonstrated that the trail-limb knee extensor moment is greater when the lead limb is more vertical, a likely consequence of the trail limb needing to support a greater percentage of the total body weight when the lead shank angle is constrained to a vertical position (ie, the lead knee is restricted in its anterior displacement with respect to the ankle joint).[26]

An important consideration of any weight-bearing exercise focusing on strengthening the muscles of the lower extremity is patellofemoral joint reaction force and stress. Overuse related to the frequency or magnitude of load or stress (force per unit area) at this joint has been implicated in the etiology of patellofemoral pain.[2,27] In addition to the peak patellofemoral joint reaction force or stress magnitude, time-dependent variables, such as the area under the force or stress time curve, provide additional information regarding the mechanics of this joint for a given exercise. For example, a condition involving decreased load or stress over a longer period may still introduce supraphysiologic overload at the joint and disrupt soft tissue homeostasis.[10] Patellofemoral joint reaction forces and stresses have often been considered in past investigations of the forward lunge, which have focused on the influence of step length, technique, or trunk posture.[13,14,17] Unilateral strength training or rehabilitation would benefit from increased information related to both limbs during the forward lunge, especially considering that this exercise is often included in strength-training protocols for the rehabilitation of patellofemoral pain.[3,23,29]

The objective of this study was to describe patellofemoral kinetics of both the lead and trail limbs during 3 variations of the forward lunge exercise, each imposing a different trunk position and/or lead-limb shank angle. It was hypothesized that knee extensor moment, patellofemoral joint reaction force, and patellofemoral joint stress would differ between the lead limb and the trail limb, and also vary as a function of trunk and lead-limb positions.

METHODS

Participants

Eighteen healthy, recreationally active participants (5 female, 13 male; mean ± SD age, 34 ± 13 years; weight, 75.9 ± 10.7 kg; height, 176.4 ± 8.9 cm) provided informed consent prior to participation. All participants were free of lower extremity pain, and most were familiar with the forward lunge and its variations. However, some of the variations investigated in the present study were novel to some participants. The protocols and procedures of the current study were approved by the Institutional Review Board of the University of Massachusetts-Amherst.

Lunge Variations

Three lunge variations, each named according to positions of the trunk and shank (**FIGURE 1**), were examined in this study. The first variation was termed "vertical" (VV). During VV lunges, both the trunk and shank were maintained in a vertical position throughout the duration of the lunge when the lead limb was in contact with the ground. Specifically, the spine was neutral (ie, natural lumbar lordosis) and perpendicular to the ground. The shank also was perpendicular to the ground, which was achieved by maintaining a neutral ankle angle; thus, anterior translation of the lead-limb knee joint center over the ankle joint center was prevented. The second variation was termed "forward" (FF). The FF variation involved forward translation of the trunk and shank. Specifically, the forward translation of the trunk was achieved by maximally flexing the hip while maintaining a neutral spine, and that of the shank by maximally dorsiflexing the ankle with the heel remaining on the ground. This produced anterior translation of the lead-limb knee joint center with respect to the ankle joint center, the extent of which was dependent on the individual's dorsiflexion range of motion. The third variation was termed "forward vertical" (FV). During FV lunges, the trunk underwent forward translation and the shank remained vertical. Again, the forward translation of the trunk was achieved through flexion at the hip, while maintaining a neutral spine, and the vertical shank was maintained by 0° of ankle dorsiflexion.

Each lunge variation was performed with body weight as the only resistance. For all trials, participants were instructed to lunge to a depth just prior to the trail knee touching the ground. At least 1 of the investigators was present to provide coaching and instruction when participants were practicing the lunge varieties.

Stride length for all participants was normalized to limb length (mean ± SD, 84.4 ± 5.6 cm). Participants were provided a target landing area ranging from 75% to 100% of limb length and were instructed to strike with the heel at a preferred stride length, provided that it was within the target area. All lunges were temporally regulated by a metronome set at 55 beats per minute. Participants were instructed to step forward to begin the lunge on the first beat, push back to return to standing position on the third beat, and begin the next repetition on the fifth beat. Thus, each repetition consisted of approximately 2 seconds of forward

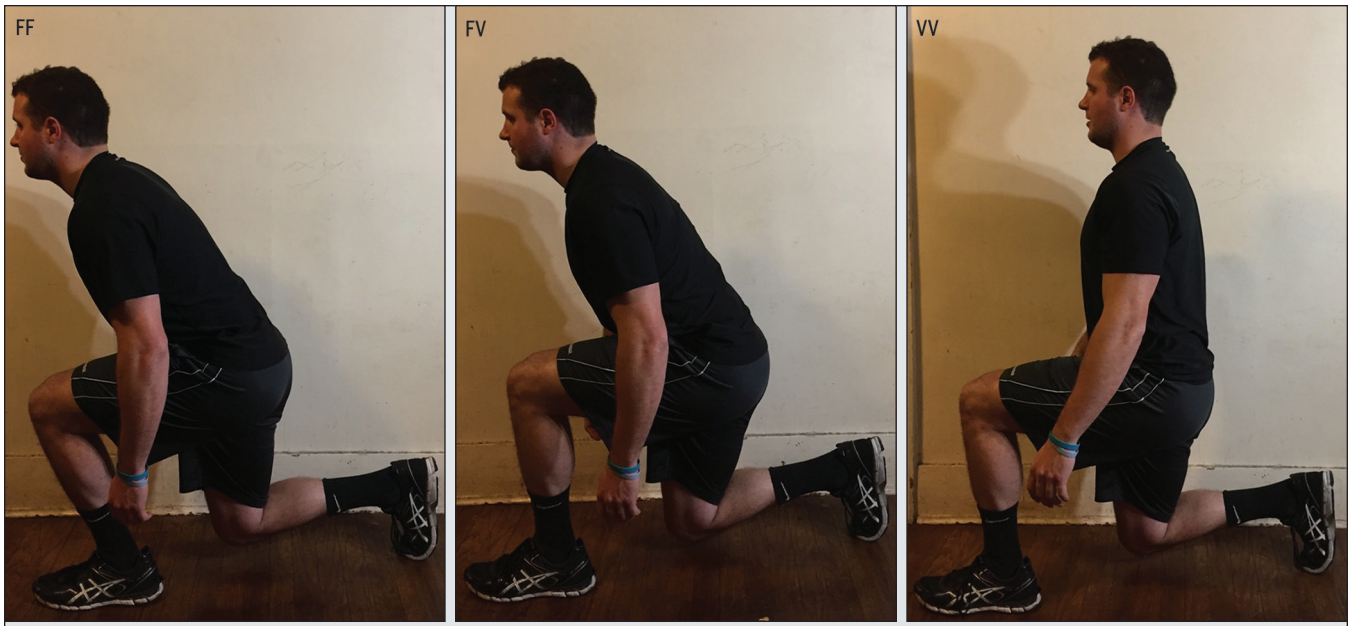


FIGURE 1. Illustration of the 3 different lunge varieties. The first and second letters (eg, F for forward, V for vertical) represent the position of the trunk and shank, respectively. Abbreviations: FF, forward trunk, forward shank; FV, forward trunk, vertical shank; VV, vertical trunk, vertical shank.

lunging and 2 seconds of pushing back to the standing position.

Data Collection

All participants attended 2 data-collection sessions a week apart. For both sessions, participants were instructed to wear athletic clothing and footwear. During session 1, participants were taught how to perform the 3 lunging techniques and were familiarized with the timing and stride-length targets as described above. First, study participants were instructed on how to perform the 3 lunge variations with appropriate form, timing, and stride length, while emphasizing the positions of the trunk and shank for each technique. Next, participants were asked to perform each lunge variation, and the instructional cues described above were generally successful in achieving proper form. By the end of session 1, all participants could independently perform the 3 lunge techniques with either the left or right leg serving as the lead limb.

Session 2 began with recording each participant's age, height, weight, and limb length (as measured from the lateral malleolus to the greater trochanter). Next, reflective markers were placed on the participant for motion capture. A set of unilateral, passive, retroreflective spherical markers was used to assess the lower extremity, which consisted of marker clusters attached to the shank and thigh by athletic tape, and individual markers on the right and left anterior superior iliac spine, sacrum, lateral and medial epicondyles of the knee, lateral and medial malleoli, calcaneus, and second and fifth metatarsal heads. Markers were placed on the right limb, which was the preferred limb (defined as the limb with which the subject would kick a ball) for all members of this sample population.

Each subject was then instructed to perform 6 different trials of lunges (3 lunge variations for both limbs). Each trial consisted of 6 repetitions, with data collection beginning after the completion of the second repetition. The order of trials was counterbalanced between participants. A 2-minute rest was provided between each trial, and a 5-minute break was provided at the halfway point of the experiment to mitigate the effects of fatigue. All trials were visually inspected to ensure that the lunges were of appropriate body position, timing, and depth. If technique, stride length, or timing was improper, the trial was eliminated and repeated.

The same limb (ie, the right limb) was analyzed for all trials; however, that limb's role varied as either the lead or trail limb, depending on the experimental condition. Thus, during lead-limb trials, participants lunged onto the force plate by leading with their right limb, and during trail-limb trials, they started by stepping off the force plate and lunged onto their left limb.

Kinematic data were captured at 100 Hz using 12 infrared cameras (OptiTrack; NaturalPoint, Inc, Corvallis, OR). Ground reaction forces were sampled at 500 Hz with an AMTI force plate (BP400600; Advanced Mechanical Technology, Inc, Watertown, MA) and temporally synchronized with the kinematic data by Capture2D software (Version 1.0.0.49; C-Motion, Inc, Germantown, MD).

Data Analysis

All data analyses were performed with Visual3D software (Version 4.92.0; C-Motion, Inc) and custom MATLAB scripts (The MathWorks, Inc, Natick, MA). Kinematic data were filtered (low-pass Butterworth filter; cutoff, 6 Hz) and inter-

TABLE Knee Joint Impulses and Patellofemoral Mechanics of the Lead and Trail Limbs by Lunge Type Across All Participants*

Measure/Limb	FF	FV	VV
Trial duration, s			
Lead	2.2 0.3	2.3 0.3	2.0 0.3
Trail	3.6 0.3	3.7 0.2	3.5 0.3
Knee extensor impulse, Nm·s·kg^{-1}			
Lead	1.8 0.4†	1.5 0.5	1.5 0.3
Trail	1.9 0.6	2.6 0.5‡§	2.8 0.5‡§
PFJ RF maximum, N·kg^{-1}			
Lead	42.5 8.0§‖	35.4 7.5	40.7 10.0
Trail	32.6 10.4	43.8 10.6‡§	50.7 8.1‡§‖
PFJ RF impulse, N·s·kg^{-1}			
Lead	58.9 11.8†‖	49.1 15.3	48.5 14.4
Trail	58.3 21.8	81.7 19.4‡§	90.6 15.4‡§‖
PFS maximum, MPa·kg^{-1}			
Lead	0.0945 0.02‖	0.0840 0.02	0.0967 0.02‖
Trail	0.0940 0.03	0.113 0.03‡§	0.125 0.02‡§‖
PFS impulse, MPa·s·kg^{-1}			
Lead	0.133 0.03	0.116 0.03	0.114 0.03
Trail	0.174 0.06§	0.222 0.04‡§	0.245 0.04‡§

Abbreviations: FF, forward trunk, forward shank; FV, forward trunk, vertical shank; PFJ RF, patellofemoral joint reaction force; PFS, patellofemoral stress; VV, vertical trunk, vertical shank.
**Values are mean ± SD.*
†Greater than VV within the same limb.
‡Greater than FF within the same limb.
§Greater than the other limb within same posture (P<.01).
‖Greater than FV within the same limb.

polated with a third-order polynomial. Inverse dynamics equations, incorporating segmental inertial properties,[9] were used to quantify the knee extensor moments for each limb (lead or trail) and lunge condition. Patellofemoral joint reaction force was estimated as a function of knee angle and knee extensor moment, using methods described previously.[6] Patellofemoral contact area was estimated as a linearly increasing function of knee angle from 0° to 90° of flexion, as previously described[14] and adapted from past magnetic resonance imaging studies.[25] For knee flexion angles from 90° to 120°, a linear 5% decrease in contact area was assumed, as described previously.[21] Patellofemoral joint stress was predicted by dividing the estimated patellofemoral joint reaction force by the estimated contact area.

Different criteria were utilized to define the duration of the lunge for both the lead and trail limbs. For the lead limb, the trial was considered finished when the ground reaction force vector was greater than 10 N. Because there is no flight phase for the trail limb, the lunge was considered to take place when the center of pressure under the trail foot was lateral to the medial malleolus in the frontal plane. This resulted in a trail-limb lunge duration commencing briefly after the lead-limb step and concluding briefly after the lead limb had returned to its initial position.

Grand means for all variables were calculated for each experimental condition as the average of repetitions for all 18 participants. Peak values were determined for patellofemoral joint reaction force and stress. Knee extensor moment impulse, patellofemoral joint reaction force impulse, and patellofemoral joint stress impulse were quantified using a trapezoidal method to approximate the area under the time-series curve throughout the exercise. All data were normalized to body weight. Time-series data were normalized as a percentage of stance, as per the aforementioned prescribed definitions, although all impulse values were calculated using non–time-normalized data.

Statistical Analysis

A 2-by-3 (limb-by-lunge variation) repeated-measures analysis of variance was performed for each of 5 outcome variables (knee extensor impulse, peak patellofemoral joint reaction force, patellofemoral joint reaction force impulse, peak patellofemoral joint stress, patellofemoral joint stress impulse). Paired *t* tests were used for post hoc comparisons (ie, comparing the 3 lunge variations intralimb, and lead limb versus trail limb comparisons for each lunge variation). Hypothesis testing was performed with a Bonferroni-corrected alpha of .01.

RESULTS

Lunge Characteristics

The differences in loading duration between the lead and trail limbs are presented in the **TABLE**. On average, the peak knee flexion angle on the lead limb was 90° in the vertical shank positions by design, and approximately 100° during the shank-forward position (**FIGURE 2**). Although the trail limb kinematics were not controlled, both vertical-shank conditions resulted in a peak knee flexion angle of approximately 100°, while the shank-forward condition resulted in a peak knee flexion angle of approximately 105° (**FIGURE 2**).

Knee Extensor Impulse

There was a significant main effect for limb ($P<.0001$) and posture ($P<.0001$), and a significant limb-by-posture interaction ($P<.0001$) (**TABLE**) for lead-limb knee extensor impulse (**FIGURE 3**). Post hoc testing revealed that the lead-limb knee extensor impulse was significantly greater during FF relative to VV ($P = .0005$). For the trail limb, knee extensor impulse was significantly less during FF trials compared to both FV ($P<.0001$) and VV ($P<.0001$). During FV and VV variations, the trail-limb knee extensor impulse was significantly greater than that of the lead limb (both, $P<.0001$). There was no difference between the lead and trail limbs during FF ($P = .57$).

Patellofemoral Joint Reaction Forces

For peak patellofemoral joint reaction force, there was a significant main effect for posture ($P<.0001$), and a significant limb-by-posture interaction ($P<.0001$). There was no significant main effect for limb ($P = .1360$) (**TABLE**). Post hoc testing revealed that, in the lead limb, peak force was significantly greater during FF relative to FV ($P = .0001$); in the trail limb, all 3 postures were significantly different from each other ($P<.0001$), with the greatest patellofemoral joint reaction force occurring during VV (**FIGURES 4** and **5**). Peak patellofemoral joint reaction force was significantly greater in the trail limb compared to the lead limb for FV ($P = .0021$) and VV ($P = .0005$), while it was significantly greater in the lead limb compared to the trail limb during FF ($P = .0046$).

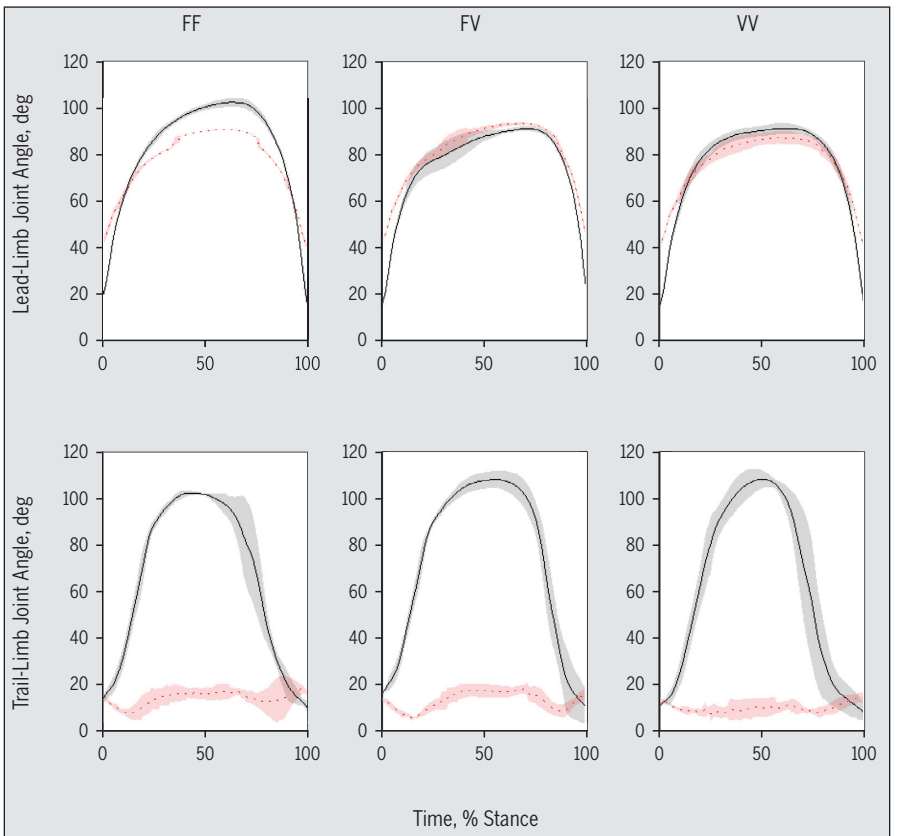


FIGURE 2. Lead-limb and trail-limb flexion angle (SD) (solid lines, knee; dashed lines, hip) versus time for the 3 experimental conditions. Abbreviations: FF, forward trunk, forward shank; FV, forward trunk, vertical shank; VV, vertical trunk, vertical shank.

With respect to patellofemoral joint reaction force, there was a significant main effect for limb ($P<.0001$), posture ($P<.0001$), and limb-by-posture interaction ($P<.0001$). Post hoc testing revealed that, in the lead limb, FF was significantly greater than both FV ($P = .0006$) and VV ($P = .0002$). In the trail limb, FF was significantly less than both FV ($P<.0001$) and VV ($P<.0001$). The trail limb was greater than the lead limb for FV ($P<.0001$) and VV ($P<.0001$), and not significantly different during FF ($P = .93$).

Patellofemoral Joint Stress

There was a significant main effect for limb ($P = .0020$), posture ($P<.0001$), and limb-by-posture interaction ($P<.0001$) (**TABLE**) for peak patellofemoral joint stress (**FIGURES 6** and **7**). Post hoc testing revealed that in the lead limb, peak stress during FV was significantly less than during both FF ($P = .0006$) and VV ($P = .0016$). In the trail limb, FF was less than FV ($P = .0002$) and VV ($P<.0001$). In addition, FV was less than VV ($P = .0012$). Peak patellofemoral joint stress was greater in the trail limb compared to the lead limb during FV and VV ($P<.0001$), while not significantly different in FF ($P = .95$).

The analysis of variance comparing patellofemoral joint stress impulse revealed a significant main effect for limb ($P<.0001$), posture ($P<.0001$), and limb-by-posture interaction ($P<.0001$). Post hoc testing revealed that, in the lead limb, FF was not significantly greater than either VV ($P = .0020$) or FV ($P = .0035$).

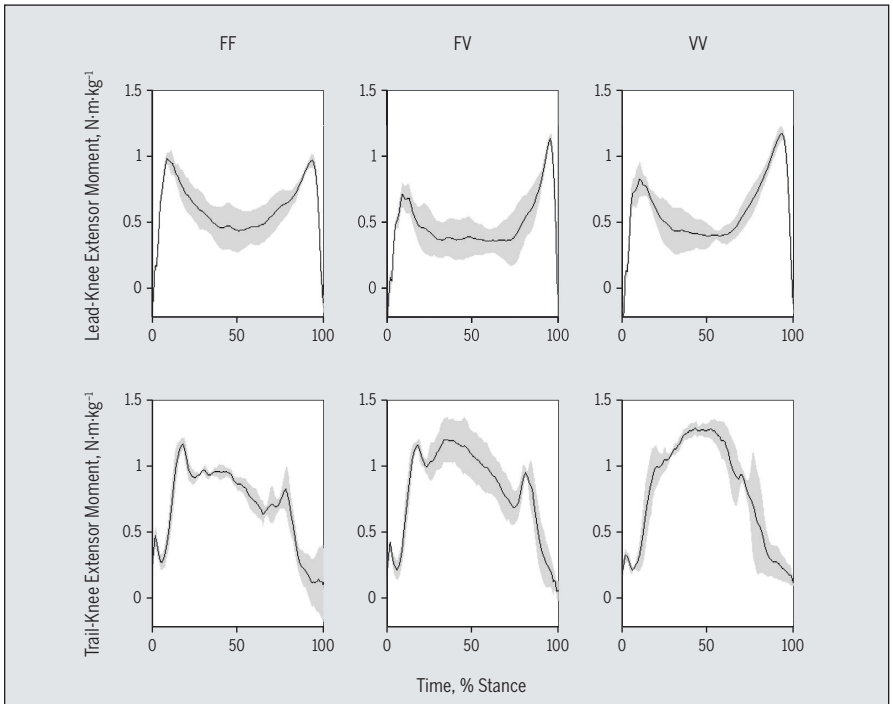


FIGURE 3. Lead-limb and trail-limb knee extensor moment (SD) versus time for the 3 experimental conditions. Abbreviations: FF, forward trunk, forward shank; FV, forward trunk, vertical shank; VV, vertical trunk, vertical shank.

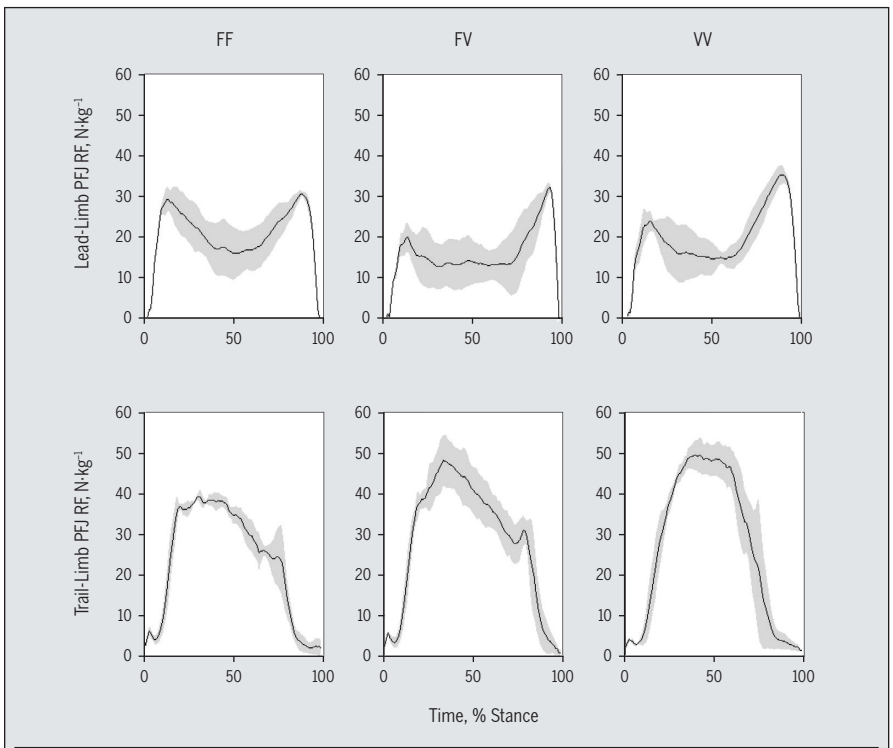


FIGURE 4. Lead-limb and trail-limb patellofemoral joint reaction force (SD; positive values indicate compressive force) versus time for the 3 experimental conditions. Abbreviations: FF, forward trunk, forward shank; FV, forward trunk, vertical shank; PFJ RF, patellofemoral joint reaction force; VV, vertical trunk, vertical shank.

In the trail limb, the FF posture was significantly less than both FV (P = .0001) and VV (P<.0001). Patellofemoral joint stress impulse was greater in the trail limb compared to the lead limb during FV (P<.0001), VV (P<.0001), and FF (P = .0090).

DISCUSSION

The present study quantified the effects of altering shank angle and trunk position on the biomechanics of both the lead and trail limbs during the forward lunge exercise. Consistent with the hypothesis, our data suggest that both joint moments and patellofemoral kinetics of the lead and trail limbs are significantly influenced by lunge technique. Considering that the lunge is a bilateral exercise, evidence-based recommendations in a rehabilitative or performance setting should consider not just the lead limb, as has been the focus of past research, but also the trail limb.

The initial weight-acceptance phase of the lead limb is characterized by eccentric loading of the knee extensor musculature, followed by concentric loading during the propulsive phase. When the shank was forward, lead-limb knee extensor moments were greater relative to both vertical-shank conditions. The opposite trend was evident for the trail limb, with the lowest knee extensor moments occurring when the lead shank was forward. A post hoc analysis of vertical ground reaction force magnitude revealed that FF resulted in an approximately 15% increase for the lead limb and an approximately 30% decrease for the trail limb relative to FV. This suggests that, in addition to altering the ranges of motion in the lower extremities, lead-limb shank position also will influence the load distribution between the limbs. Similar outcomes have been demonstrated in split squats, another bilateral ground-based exercise.[26]

Previous research on the forward lunge has demonstrated that forward lean of the trunk results in greater recruitment of the hamstrings.[17] Past investigations also have

elucidated the role of changing trunk position on lower extremity musculature, specifically that forward lean of the trunk increases ankle plantar flexor and hip extensor joint moments.[17] Taken together, our results suggest that the demand to the knee extensor musculature, and likely all muscle groups of the limbs, will change as a function of lunge technique and that these changes are limb dependent.

Our data suggest that a vertical shank reduces peak patellofemoral joint reaction force and stress on the lead limb, but at the expense of increased patellofemoral joint loading in the trail limb. These findings are consistent with previous results suggesting that a greater percentage of body weight is supported by the lead limb with increasing shank angles (ie, more anterior displacement of the knee joint relative to the ankle) during the split squat exercise.[26] This asymmetrical loading of the patellofemoral joints is an important consideration, given that the trail limb is supporting body weight for a longer duration (approximately 3.5 seconds) than the lead limb (approximately 2.0 seconds) during all lunge techniques studied.

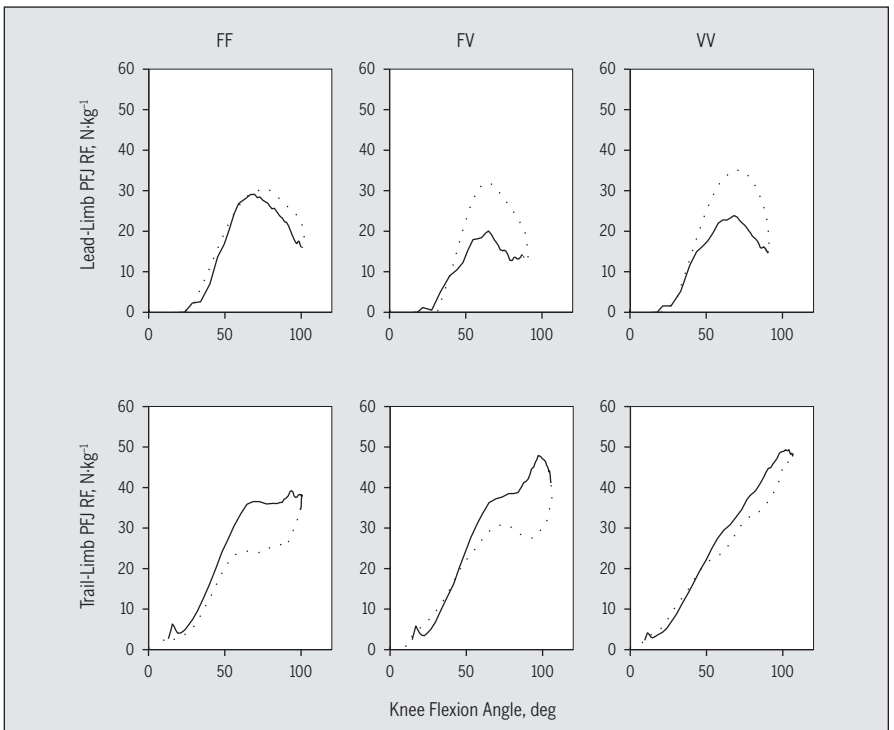


FIGURE 5. Lead-limb and trail-limb patellofemoral joint reaction force (solid lines, 0%-50% of stance; dashed lines, 50%-100% of stance) versus knee angle for the 3 experimental conditions. Abbreviations: FF, forward trunk, forward shank; FV, forward trunk, vertical shank; PFJ RF, patellofemoral joint reaction force; VV, vertical trunk, vertical shank.

Peak loading of the lead-limb patellofemoral joint (approximately 35-42 $N \cdot kg^{-1}$) was found to be generally consistent with past research for a variety of forward lunge techniques.[13,14,16,17] In addition, peak loading of the trail limb was found to be greater than that of the lead limb during postures in which the shank was maintained in a vertical position (approximately 44-51 $N \cdot kg^{-1}$). These findings illustrate that peak patellofemoral joint reaction force in both limbs is relatively high for the lunge exercise relative to activities such as walking (9.5 $N \cdot kg^{-1}$)[6] and cycling (approximately 18 $N \cdot kg^{-1}$),[24] while comparable to others, including bilateral barbell squatting with 35% of body weight (34.4 $N \cdot kg^{-1}$),[28] stair climbing (30-37 $N \cdot kg^{-1}$),[5,8] running (approximately 55 $N \cdot kg^{-1}$),[18] step-ups (43.6 $N \cdot kg^{-1}$),[7] and step-downs (51.1 $N \cdot kg^{-1}$).[7]

Considering the progressive increase in patellofemoral joint reaction force and stress from initial contact to midstance, our data are consistent with the recommendations of others[14] that working through a limited range of motion (specifically, less than 60° of knee flexion) may be necessary to reduce peak patellofemoral joint reaction force in the lead limb during the forward lunge. However, if the primary concern when performing a forward lunge in a rehabilitative setting is not patellofemoral joint force, these recommendations may not be warranted. For example, if the aim is to reduce strain on the anterior cruciate ligament, past experimental data[16,20] and computer simulations[1] suggest minimal loading of the anterior cruciate ligament during the forward lunge exercise.

It is not currently known what magnitudes of patellofemoral joint stress are of concern clinically; however, the values in the present study can be compared to other activities in the literature. Past investigations have most often reported patellofemoral joint stress as stress magnitude, rather than as stress normalized to body weight. If the normalized stress values in the present study are converted, the magnitudes (approximately 7-9 MPa) are generally consistent with past lunge research investigating forward-shank (approximately 7 MPa) and vertical-shank (approximately 5 MPa) lunges,[14] wall squats (approximately 9 MPa),[15] and stair ascent (approximately 7 MPa).[5] All values of patellofemoral joint reaction force and stress display generally consistent trends. As such, any recommendations regarding lunge technique would not be altered whether one were to use either criterion as a means for exercise selection.

It is important to acknowledge the limitations of the present study. A brief analysis of frontal and transverse plane kinematics of the hip, knee, and ankle revealed generally similar values across the lunge variations in our sample, and thus, this analysis was restricted to the sagittal plane. The predictions of the quadriceps

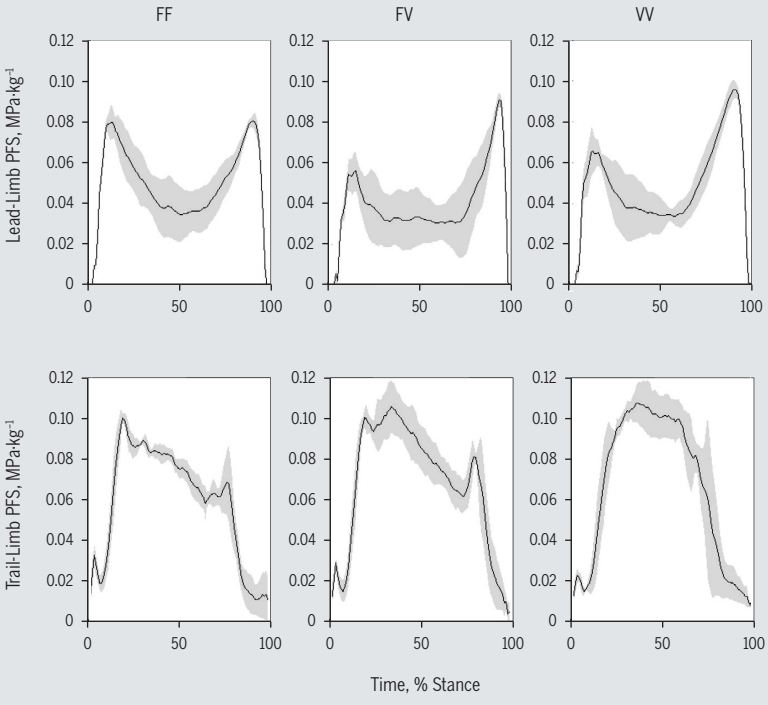


FIGURE 6. Lead-limb and trail-limb patellofemoral joint stress (SD) versus time for the 3 experimental conditions. Abbreviations: FF, forward trunk, forward shank; FV, forward trunk, vertical shank; PFS, patellofemoral stress; VV, vertical trunk, vertical shank.

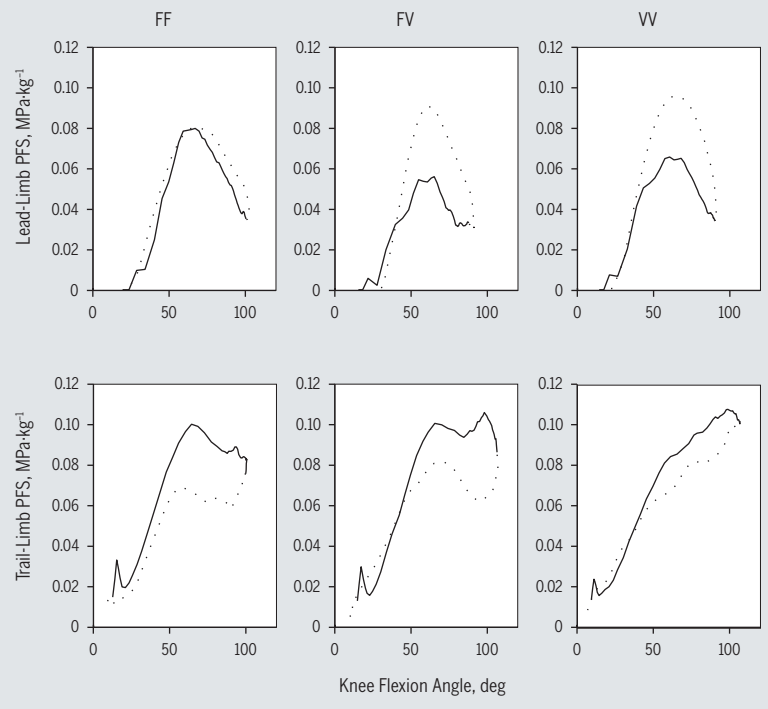


FIGURE 7. Lead-limb and trail-limb patellofemoral joint stress (solid lines, 0%-50% of stance; dashed lines, 50%-100% of stance) versus knee angle for the 3 experimental conditions. Abbreviations: FF, forward trunk, forward shank; FV, forward trunk, vertical shank; PFS, patellofemoral stress; VV, vertical trunk, vertical shank.

tendon moment arm and the patellofemoral contact area were not subject specific; however, this concern was mitigated by the repeated-measures design of the experiment. The model in this investigation utilized net knee joint moment calculated from an inverse dynamics analysis to estimate quadriceps tendon force. As a result, antagonist cocontraction from the knee flexors was not taken into consideration. In activities likely to involve moderate to high activation of muscles such as the gastrocnemius and the hamstrings, this would result in an underestimation of quadriceps tendon force.

Alternative techniques incorporate static optimization criteria as additional inputs to predict muscle forces from net joint moments. Past research has demonstrated a 54% difference in patellofemoral joint stress between models that consider solely net joint moment and models that include static optimization during the bilateral barbell squat exercise.[22] It is important to note that musculoskeletal models based on static optimization are not without limitations, as these models are often based on assumptions regarding the selection of appropriate optimization functions, and are based on limited experimental data. Regardless, it could be speculated that the patellofemoral joint stresses of the lunge varieties in the present investigation were underestimated. Electromyographic assessment in previous research has demonstrated that forward trunk lean (much greater than that analyzed in the present study) compared to an upright trunk during the forward lunge generally results in similar, although with statistically significant differences, biceps femoris (trunk vertical, 11.9% maximum voluntary contraction [MVC]; trunk forward, 17.9% MVC) and gluteus maximus muscle activation (trunk vertical, 18.5% MVC; trunk forward, 22.3% MVC), with no difference in vastus lateralis activation (trunk vertical, 45.6% MVC; trunk forward, 50.9% MVC).[15] Therefore, it is suggested that muscle activation patterns are generally similar across the lunge variations for the

lead limb, especially for the quadriceps muscle group. In addition, our predictions of lead-limb patellofemoral joint stresses with lunges in the vertical and forward shank positions are generally consistent in trend and magnitude with those investigated in past research using a more sophisticated musculoskeletal model that included predictions of all lower extremity muscle forces.[14]

CONCLUSION

Patellofemoral joint stresses in the trail limb were greater than stresses in the lead limb. In addition, joint moments and patellofemoral joint reaction force and stress in each limb were significantly altered as a result of changing trunk and shank position. Specifically, these findings suggest that a vertical trunk and shank position will result in significant stress at the patellofemoral joint in the trail limb. It is recommended that the stresses imposed on both limbs during the forward lunge be a consideration during all cases of preventative or rehabilitative prescription of the forward lunge exercise. ◉

KEY POINTS

FINDINGS: Patellofemoral joint stresses in the trail limb were greater than those in the lead limb during many of the forward lunge postures under investigation. Peak patellofemoral joint stresses in the trail knee occurred when the trunk and lead shank were both vertical.

IMPLICATIONS: The prescription of the forward lunge in an exercise or rehabilitative setting should consider the stresses imposed on both the lead and trail knees.

CAUTION: The quadriceps muscle force prediction does not consider cocontraction of antagonists at the knee, and may result in underestimation of patellofemoral joint stresses. Care should be taken when extrapolating these findings to an athletic population, or a population well trained in the forward lunge exercise.

ACKNOWLEDGMENTS: *The authors wish to acknowledge the technical support of Cybex International, Inc.*

REFERENCES

1. Alkjaer T, Wieland MR, Andersen MS, Simonsen EB, Rasmussen J. Computational modeling of a forward lunge: towards a better understanding of the function of the cruciate ligaments. *J Anat*. 2012;221:590-597. https://doi.org/10.1111/j.1469-7580.2012.01569.x
2. Besier TF, Fredericson M, Gold GE, Beaupré GS, Delp SL. Knee muscle forces during walking and running in patellofemoral pain patients and pain-free controls. *J Biomech*. 2009;42:898-905. https://doi.org/10.1016/j.jbiomech.2009.01.032
3. Boling MC, Bolgla LA, Mattacola CG, Uhl TL, Hosey RG. Outcomes of a weight-bearing rehabilitation program for patients diagnosed with patellofemoral pain syndrome. *Arch Phys Med Rehabil*. 2006;87:1428-1435. https://doi.org/10.1016/j.apmr.2006.07.264
4. Boudreau SN, Dwyer MK, Mattacola CG, Lattermann C, Uhl TL, McKeon JM. Hip-muscle activation during the lunge, single-leg squat, and step-up-and-over exercises. *J Sport Rehabil*. 2009;18:91-103. https://doi.org/10.1123/jsr.18.1.91
5. Brechter JH, Powers CM. Patellofemoral joint stress during stair ascent and descent in persons with and without patellofemoral pain. *Gait Posture*. 2002;16:115-123. https://doi.org/10.1016/S0966-6362(02)00090-5
6. Brechter JH, Powers CM. Patellofemoral stress during walking in persons with and without patellofemoral pain. *Med Sci Sports Exerc*. 2002;34:1582-1593.
7. Chinkulprasert C, Vachalathiti R, Powers CM. Patellofemoral joint forces and stress during forward step-up, lateral step-up, and forward step-down exercises. *J Orthop Sports Phys Ther*. 2011;41:241-248. https://doi.org/10.2519/jospt.2011.3408
8. Costigan PA, Deluzio KJ, Wyss UP. Knee and hip kinetics during normal stair climbing. *Gait Posture*. 2002;16:31-37. https://doi.org/10.1016/S0966-6362(01)00201-6
9. Dempster WT. Space Requirements of the Seated Operator: Geometrical, Kinematic, and Mechanical Aspects of the Body With Special Reference to the Limbs. Ann Arbor, MI: University of Michigan; 1955.
10. Dye SF. The pathophysiology of patellofemoral pain: a tissue homeostasis perspective. *Clin Orthop Relat Res*. 2005:100-110.
11. Ebben WP, Feldmann CR, Dayne A, Mitsche D, Alexander P, Knetzger KJ. Muscle activation during lower body resistance training. *Int J Sports Med*. 2009;30:1-8. https://doi.org/10.1055/s-2008-1038785
12. Ekstrom RA, Donatelli RA, Carp KC. Electromyographic analysis of core trunk, hip, and thigh muscles during 9 rehabilitation exercises. *J Orthop Sports Phys Ther*. 2007;37:754-762. https://doi.org/10.2519/jospt.2007.2471
13. Escamilla RF, Zheng N, MacLeod TD, et al. Patellofemoral compressive force and stress during the forward and side lunges with and without a stride. *Clin Biomech (Bristol, Avon)*. 2008;23:1026-1037. https://doi.org/10.1016/j.clinbiomech.2008.05.002
14. Escamilla RF, Zheng N, MacLeod TD, et al. Patellofemoral joint force and stress between a short- and long-step forward lunge. *J Orthop Sports Phys Ther*. 2008;38:681-690. https://doi.org/10.2519/jospt.2008.2694
15. Escamilla RF, Zheng N, MacLeod TD, et al. Patellofemoral joint force and stress during the wall squat and one-leg squat. *Med Sci Sports Exerc*. 2009;41:879-888. https://doi.org/10.1249/MSS.0b013e31818e7ead
16. Escamilla RF, Zheng N, MacLeod TD, et al. Cruciate ligament forces between short-step and long-step forward lunge. *Med Sci Sports Exerc*. 2010;42:1932-1942. https://doi.org/10.1249/MSS.0b013e3181d966d4
17. Farrokhi S, Pollard CD, Souza RB, Chen YJ, Reischl S, Powers CM. Trunk position influences the kinematics, kinetics, and muscle activity of the lead lower extremity during the forward lunge exercise. *J Orthop Sports Phys Ther*. 2008;38:403-409. https://doi.org/10.2519/jospt.2008.2634
18. Flynn TW, Soutas-Little RW. Patellofemoral joint compressive forces in forward and backward running. *J Orthop Sports Phys Ther*. 1995;21:277-282. https://doi.org/10.2519/jospt.1995.21.5.277
19. Fry AC, Smith JC, Schilling BK. Effect of knee position on hip and knee torques during the barbell squat. *J Strength Cond Res*. 2003;17:629-633.
20. Heijne A, Fleming BC, Renstrom PA, Peura GD, Beynnon BD, Werner S. Strain on the anterior cruciate ligament during closed kinetic chain exercises. *Med Sci Sports Exerc*. 2004;36:935-941.
21. Huberti HH, Hayes WC. Patellofemoral contact pressures. The influence of Q-angle and tendofemoral contact. *J Bone Joint Surg Am*. 1984;66:715-724.
22. Kernozek TW, Vannatta CN, van den Bogert AJ. Comparison of two methods of determining patellofemoral joint stress during dynamic activities. *Gait Posture*. 2015;42:218-222. https://doi.org/10.1016/j.gaitpost.2015.05.017
23. Natri A, Kannus P, Järvinen M. Which factors predict the long-term outcome in chronic patellofemoral pain syndrome? A 7-yr prospective follow-up study. *Med Sci Sports Exerc*. 1998;30:1572-1577.
24. Neptune RR, Kautz SA. Knee joint loading in forward versus backward

pedaling: implications for rehabilitation strategies. *Clin Biomech (Bristol, Avon)*. 2000;15:528-535. https://doi.org/10.1016/S0268-0033(00)00005-X

25. Salsich GB, Ward SR, Terk MR, Powers CM. In vivo assessment of patellofemoral joint contact area in individuals who are pain free. *Clin Orthop Relat Res*. 2003:277-284.
26. Schütz P, List R, Zemp R, Schellenberg F, Taylor WR, Lorenzetti S. Joint angles of the ankle, knee, and hip and loading conditions during split squats. *J Appl Biomech*. 2014;30:373-380. https://doi.org/10.1123/jab.2013-0175
27. Thomeé R, Augustsson J, Karlsson J. Patellofemoral pain syndrome: a review of current issues. *Sports Med*. 1999;28:245-262.
28. Wallace DA, Salem GJ, Salinas R, Powers CM. Patellofemoral joint kinetics while squatting with and without an external load. *J Orthop Sports Phys Ther*. 2002;32:141-148. https://doi.org/10.2519/jospt.2002.32.4.141
29. Witvrouw E, Danneels L, Van Tiggelen D, Willems TM, Cambier D. Open versus closed kinetic chain exercises in patellofemoral pain: a 5-year prospective randomized study. *Am J Sports Med*. 2004;32:1122-1130. https://doi.org/10.1177/0363546503262187
30. Zatsiorsky V, Seluyanov V. The mass and inertia characteristics of the main segments of the human body. In: Matsui H, Kobayashi K, eds. *Biomechanics VIII-B. Proceedings of the Eighth International Congress of Biomechanics*. Champaign, IL: Human Kinetics; 1983:1152-1159.

MORE INFORMATION
WWW.JOSPT.ORG

BROWSE Collections of Articles on *JOSPT*'s Website

JOSPT's website (**www.jospt.org**) offers readers the opportunity to browse published articles by **Previous Issues** with accompanying volume and issue numbers, date of publication, and page range; the table of contents of the **Upcoming Issue**; a list of available accepted **Ahead of Print** articles; and a listing of **Categories** and their associated article collections by type of article (Research Report, Case Report, etc).

Features further curates 3 primary *JOSPT* article collections: Musculoskeletal Imaging, Clinical Practice Guidelines, and Perspectives for Patients, and provides a directory of Special Reports published by *JOSPT*.